

Assignment 3: Raw Socket Packet Analysis and Protocol Investigation

Total Marks: 100

Objective

The objective of this assignment is to understand raw socket programming, IP packet structure, protocol headers, and packet analysis using Linux and Wireshark. Students will write a C program using raw sockets, generate protocol-specific traffic, capture packets, compare results with Wireshark, and analyze packet header fields.

Task 1: Raw Socket Packet Capture Program (25 Marks)

Develop a C program using Linux raw sockets to capture packets for one protocol based on the last digit of your roll number.

Roll Number Last Digit	Assigned Protocol
0-3	ICMP
4-6	UDP
7-9	TCP

Your program must capture and display information for at least 20 packets. For every packet, display these IP header fields:

- Source IP Address
- Destination IP Address
- Protocol Number
- TTL Value
- Packet Size

In addition, display protocol-specific fields according to your assigned protocol:

Protocol	Additional Fields
ICMP	ICMP Type and Code
UDP	Source Port and Destination Port
TCP	Source Port, Destination Port, and TCP Flags

Your program output must follow this format:

```
ROLL_NO=____
ASSIGNED_PROTOCOL=____

PACKET_NO=1
SRC_IP=____
DST_IP=____
PROTOCOL=____
PROTOCOL_NO=____
TTL=____
PACKET_SIZE=____
ADDITIONAL_FIELD=____
```

Example:

```
PACKET_NO=1
SRC_IP=10.0.0.1
DST_IP=10.0.0.2
PROTOCOL=ICMP
PROTOCOL_NO=1
```

```
TTL=64
PACKET_SIZE=84
ICMP_TYPE=8
ICMP_CODE=0
```

Task 2: Traffic Generation and Observation (15 Marks)

Generate traffic corresponding to your assigned protocol.

Protocol	Example Traffic Generation
ICMP	ping
TCP	web browsing, SSH, netcat, telnet
UDP	DNS query, iperf UDP mode, netcat UDP mode

Record the following:

1. The exact command used to generate traffic.
2. The interface on which traffic was captured.
3. How the generated traffic appeared in your raw socket program.
4. Screenshots of the terminal showing traffic generation and program output.

Example commands:

```
ping 8.8.8.8
nc -v 127.0.0.1 5000
iperf -u -c <server-ip>
```

Task 3: Wireshark Verification (20 Marks)

Capture the same traffic using Wireshark. Use an appropriate display filter, such as:

```
icmp
tcp
udp
ip.proto == 1
ip.proto == 6
ip.proto == 17
```

Select any five packets and compare the output of your raw socket program with Wireshark. Mention the Wireshark frame number for each selected packet.

Wireshark Frame No.	Source IP	Destination IP	Protocol	TTL	Packet Size	Program Output Matched?

Attach relevant Wireshark screenshots and submit the Wireshark capture file named:

```
capture.pcapng
```

Task 4: Header Analysis (20 Marks)

Choose any three captured packets and answer these questions using your observations:

1. Why is the TTL value not zero?
2. Why do packet sizes differ?
3. What information does the protocol field provide?
4. What could happen if the protocol field is modified?
5. What does the protocol-specific field reveal about the packet?

Support your answers using packet screenshots and raw socket output.

Task 5: Program Enhancement (10 Marks)

Modify your program to display one additional IP header field from the following:

- IP Version
- Header Length
- Identification
- Type of Service (ToS)
- Fragment Offset

Explain:

1. The purpose of the selected field.
2. The observed value from at least five captured packets.
3. Whether the value stayed the same or changed across packets.

Task 6: Reflection Questions (10 Marks)

Answer the following questions in your own words. Total answer length should not exceed 300 words.

1. Why are root privileges required for raw sockets?
2. How are raw sockets different from TCP/UDP sockets?
3. State one advantage and one limitation of raw sockets.
4. Mention one networking or cybersecurity application of raw sockets.

System and Execution Details

Your report must include system details. Use commands such as:

```
uname -a
gcc --version
ip addr
```

Also include the exact compile and run commands.

```
gcc raw_capture.c -o raw_capture
sudo ./raw_capture
```

Submission Guidelines

Submit one compressed folder named:

```
ROLLNO_NAME_RAWSOCKET.zip
```

The ZIP file must contain:

```
raw_capture.c
program_output.txt
capture.pcapng
screenshots/
report.pdf
```

The screenshots folder should include:

```
traffic_generation.png
program_output.png
wireshark_packets.png
comparison_packets.png
```

Report Format

1. Objective
2. System Configuration
3. Compile and Run Commands
4. Program Design
5. Traffic Generation
6. Experimental Results
7. Wireshark Verification
8. Header Analysis
9. Program Enhancement
10. Reflection Answers
11. Conclusion

Evaluation Criteria

Component	Marks
Raw socket implementation	25
Traffic generation and interpretation	15
Wireshark verification	20
Header analysis	20
Program enhancement	10
Reflection answers and report quality	10
Total Marks	100

Important Notes

1. Submission without experimental evidence, screenshots, or packet capture files will not be evaluated.

2. Students must be prepared to explain their implementation and observations during evaluation.
3. Students must capture only their own generated traffic or traffic inside a lab/Mininet environment.
4. Capturing unauthorized network traffic is strictly not allowed.
5. The assignment can be completed using help from resources, but the observations, screenshots, packet comparison, and explanation must be based on the student's own experiment.